

M11 — Activity catalog: dataclass + DB CRUD + 8 built-in activities

Phase: 7 — Activities **Status:** DONE (2026-05-26) **Blocked by:** M1 **Blocks:** M12, M14 **Estimated effort:** 1.5 days · **Actual:** 0.5 day

1. Purpose

Activities are persistent, instructor-curated case templates. The wizard's room-mode editor (M12, next) lets the operator pick an Activity for each encounter row, which then auto-fills the row's label / persona / chart-mode / scenario text. Activities also carry an optional answer key, which the cohort debrief (M14) consumes when scoring student documentation against the case-author's expected findings.

M11 ships the data layer:

1. **CRUD helpers** in `portal/ehr_db.py` for the activity table (M1 schema v4). 2. ****The Activity dataclass + 8 built-in catalog entries**** in `portal/activities.py`. The first 7 mirror `portal/data/sample_scenarios.json` 1:1 so an Activity-picked encounter and a wizard-template-picked encounter produce identical seed material. The 8th (acute respiratory failure) extends curriculum coverage to pneumonia / hypoxemia / BiPAP — a gap in the v6 samples. 3. ****seed_builtins()**** — an idempotent first-start seeder that inserts every built-in row that isn't already in the DB. Instructor edits to a built-in are NOT overwritten on re-seed.

M12 will add the HTTP routes (`/api/activities` CRUD + wizard integration); M11 is data layer only.

2. Structure

New files:

- `portal/activities.py` — Activity dataclass, `BUILTIN_ACTIVITIES`

catalog (8 entries), `seed_builtins()`, `list_all()`, `get(id)`, `to_encounter_entry(id)` convenience wrappers.

Files touched:

- `portal/ehr_db.py` — adds a "V7 — Activity catalog (M11)" section

with 5 CRUD helpers + an in-memory fallback bucket.

No server.py changes — routes land in M12.

3. Uses

- **M12 (wizard integration)** consumes:

- `ehr_db.list_activities()` for the wizard's activity picker.

- `activities.to_encounter_entry(id)` to translate a picked activity into the JSON shape `/api/room/start` accepts.

- **M14 (cohort debrief)** reads `activity.answer_key` for rubric-based scoring.

- **First-start hook** — `activities.seed_builtins()` should be

called from `_run_migrations` (or a server-startup callback). Wiring lands in M12 alongside the route surface.

- **Operator workflow:**

1. Server starts → `seed_builtins()` inserts the 8 built-ins on a fresh DB; no-op on subsequent starts. 2. Instructor edits a built-in (M12 routes) → edit persists; re- seed does NOT roll back the edit. 3. Instructor authors a new custom activity (M12) → row created with `is_builtin=False`.

4. Functions (exported API surface)

`portal/ehr_db.py` (5 new CRUD helpers)

Symbol	Signature	Purpose	
<code>create_activity</code>	<code>(*, activity_id=None, label, seed_persona_id=None, seed_modules=None, scenario_text="", default_chart_mode="shared", answer_key=None, is_builtin=False) -> dict</code>	Insert. Auto-generates <code>act_<hex10></code> id when not supplied. Returns the persisted row.	
<code>get_activity</code>	<code>`(activity_id) -> dict \</code>	<code>None`</code>	Lookup by id.
<code>list_activities</code>	<code>(*, builtin_only=False) -> list[dict]</code>	Built-ins first, then custom; both alphabetical within each group.	
<code>update_activity</code>	<code>`(activity_id, **fields) -> dict \</code>	<code>None`</code>	Patch allowed fields (<code>label / seed_persona_id / seed_modules / scenario_text / default_chart_mode / answer_key</code>). Unknown fields silently ignored (forward-compat). Returns <code>None</code> on unknown id.
<code>delete_activity</code>	<code>(activity_id) -> bool</code>	Hard delete. Built-in rows are protected — returns <code>False</code> without dropping. Idempotent (returns <code>True</code> if row was already absent).	

`portal/activities.py`

Symbol	Purpose	
Activity dataclass	Typed shape: <code>activity_id, label, seed_persona_id, seed_modules, scenario_text, default_chart_mode, answer_key, is_builtin</code> .	
<code>BUILTIN_ACTIVITIES: list[Activity]</code>	The 8-entry built-in catalog.	
<code>seed_builtins() -> int</code>	Idempotent seed; returns count of rows inserted this call.	

Symbol	Purpose	
<code>list_all() -> list[dict]</code>	Pass-through to <code>ehr_db.list_activities()</code> .	
<code>`get(activity_id) -> dict \</code>	None`	Pass-through to <code>ehr_db.get_activity</code> .
<code>`to_encounter_entry(activity_id) -> dict \</code>	None`	Translate to wizard's encounter-row JSON shape (matches <code>/api/room/start body</code>).

Built-in catalog

activity_id	label	seed_persona	modules
<code>builtin_ed_sepsis_delirium</code>	ED · Sepsis with hyperactive delirium	P-014	M32, M08, M02
<code>builtin_msurg_postop_pain</code>	Med-surg · Postop pain — RN/LPN delegation	P-012	M08, M06, M02
<code>builtin_mh_passive_si</code>	Mental health · Goals-of-care + passive SI	P-019	M39, M02
<code>builtin_substance_etoh_withdrawal</code>	Substance · Alcohol withdrawal (CIWA-Ar)	P-016	M06, M02, M39
<code>builtin_peds_febrile_child</code>	Peds · Febrile child, anxious parent	P-003	M06, M07, M03, M02
<code>builtin_geri_goals_of_care</code>	Geri · Goals-of-care with grieving family	P-013	M42, M02
<code>builtin_msurg_dka</code>	Med-surg · DKA management	P-005	M22, M06, M02
<code>builtin_msurg_resp_failure</code>	Med-surg · Acute respiratory failure	P-006	M07, M06, M02

5. Limitations

- **No HTTP surface yet.** M12 adds `/api/activities` CRUD routes

and the wizard-side picker. Until M12 lands, only the server can read/write the catalog (e.g. via a Python shell or a future bulk-import script).

- `seed_builtins()` **is not called from server startup yet.** M12

wires it into the server boot path.

- `update_activity` **patches a whitelist of fields.** Unknown

fields are silently ignored to preserve forward-compat with M12+. If a future field needs editing, add it to the allowed set.

- **No referential integrity on `seed_persona_id`.** If a future

build removes a persona from the library, activities pointing at it still exist. M12 should add a validation step at wizard finalize time (warn-but-allow).

- **Answer key shape is open-ended JSON.** No schema enforced;

M14 cohort debrief will pin down a structure when it implements scoring. Current built-ins don't ship answer keys.

- **Built-in protection is hard at the data layer.**

`delete_activity` of a built-in returns `False` — there is no "force delete" path. To reset a built-in to its pristine catalog version, M12 should add a "Reset to defaults" route that overwrites the row from `BUILTIN_ACTIVITIES`.

- **In-memory fallback uses `_mem_activities` when SQLite is unavailable.** Matches the rest of `ehr_db`'s degraded-mode contract.

6. Test status

Automated (`tests/v7/test_activity_*.py`)

Test file	Cases	Status	Last run
<code>test_activity_create_get_list_update_delete.py</code>	4 — CRUD round-trip, built-in delete protection, list ordering, bare-create defaults.	PASS	2026-05-26
<code>test_activity_seed_8_built_in_activities_present.py</code>	5 — 8 rows inserted, seed is idempotent (8 + 0 + 0), instructor edits preserved on re-seed, <code>to_encounter_entry</code> mapping, catalog covers expected curriculum areas.	PASS	2026-05-26

9/9 PASS. **Full v7 suite: 67/67 passing** (up from 58 — +9 M11 tests). **Full v6 regression on v7: 178 passed**, same 6 env-flaky pre-existing failures, **0 v7 regressions**.

Manual

None for M11 (data layer only). M12 will land the UI flow.

7. Change list

Date	Author	Change	Files
2026-05-26	claude-codre	New <code>portal/activities.py</code> with 8-entry built-in catalog + <code>seed_builtins()</code> + <code>to_encounter_entry()</code> helpers. 5 CRUD helpers in <code>ehr_db.py</code> (create/get/list/update/delete) with SQLite + in-memory fallback. 9 acceptance tests across 2 files.	<code>portal/activities.py</code> , <code>portal/ehr_db.py</code> , <code>tests/v7/test_activity_create_get_list_update_delete.py</code> , <code>tests/v7/test_activity_seed_8_built_in_activities_present.py</code>

8. Open questions / known issues

- `seed_persona_id` for the built-ins picks ONE persona from each

v6 sample's `personas[]` list — the patient. The v6 samples include 3-5 personas each (patient + family + staff). When M12 finalizes an encounter from an Activity, only the patient persona transfers. The other personas (family / staff) can be added manually in the wizard's Step 4 OR through a v7.1 "scene cast" extension.

- Built-in `scenario_text` is copied from the v6 samples

verbatim. If a sample is edited in `sample_scenarios.json`, the Activity catalog will diverge. Two options for keeping in sync: (a) load `BUILTIN_ACTIVITIES` from `sample_scenarios.json` at import time, or (b) accept the drift since Activities are meant to be editable. M11 went with (b); M12 may revisit.

- The 8th activity (`builtin_msurg_resp_failure`) is M11-authored,

not from the v6 samples. If the operator prefers the v6 sample set verbatim, they can delete it via M12 and `seed_builtins` won't re-insert it during normal operation. (It would re-insert on the NEXT fresh DB; a permanent removal requires editing `BUILTIN_ACTIVITIES` in code.)

- `answer_key` JSON shape is unconstrained. When M14 lands the cohort-debrief scorer, freeze the shape with a Pydantic model and document it [here](#).